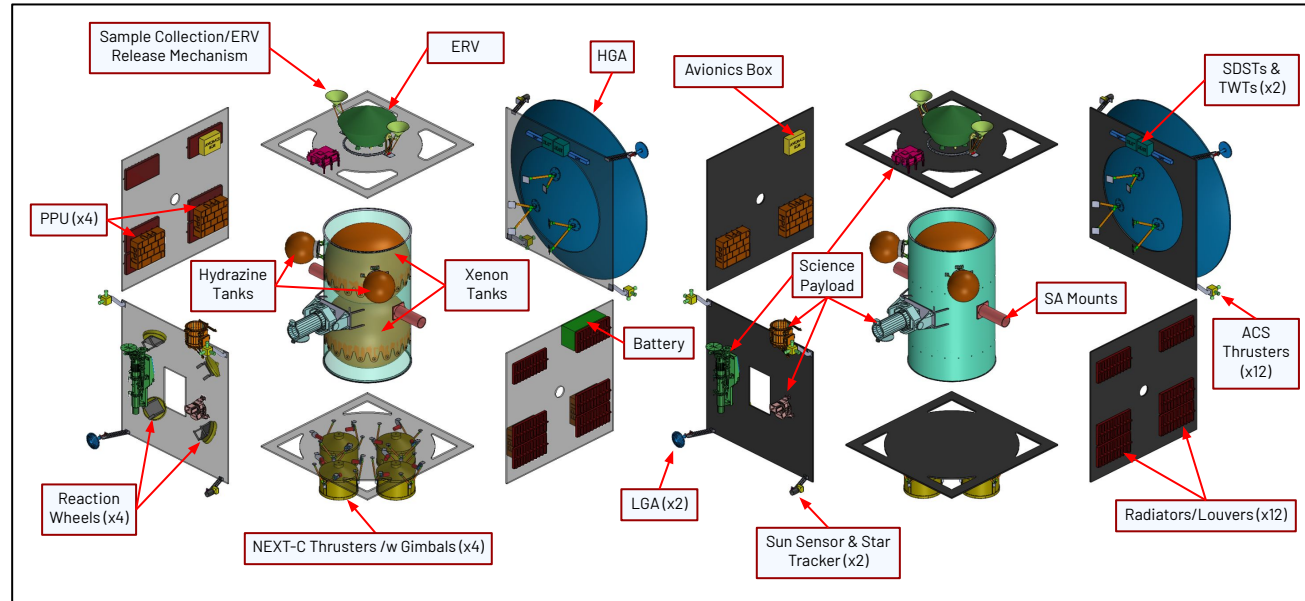
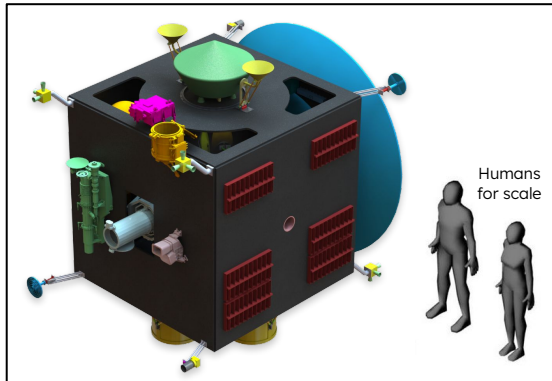
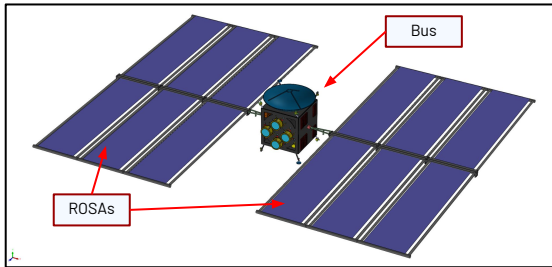


Erik Kramer | Europa Sample Return Mission Concept Study

Project → A pre-Phase A type concept study for a Europa sample return mission

Tasks → Design of **structures** subsystem, **CAD** of all spacecraft components/configurations, **systems engineering** work



Spacecraft Design and Configuration

Erik Kramer | Europa Sample Return Mission Concept Study

Objectives

- Perform a concept study for a mission to return a water plume sample from Europa
- Create a proposal with details for all subsystems showing mission viability
- Launch within 2030-2033 and have a mission duration of 10-12 years

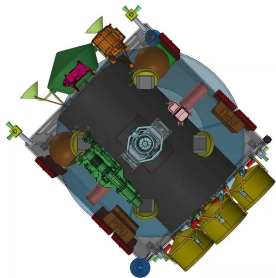
Process

- Used **SolidWorks** to design and model 100+ spacecraft components
- Tracked parts list, mass, and power through linked design spreadsheets, **MEL** and **PEL**
- Assisted other subsystems with **design iterations** and **trade studies**

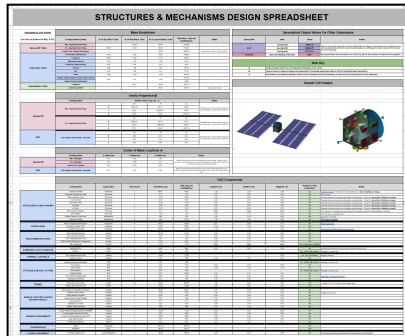
Results

- Produced a comprehensive SEP mission concept, including spacecraft design, that satisfies all level 1 to 3 requirements and lasts 11.14 years
- Collaborated to produce a 130 page report documenting the design and mission proposal

[Click](#) for Transparent Bus Rotation Animation



[Click](#) for Design Spreadsheet Example

A screenshot of a complex spreadsheet titled "STRUCTURES & MECHANISMS DESIGN SPREADSHEET". It contains multiple tabs and a large grid of data. The visible data includes columns for item numbers, descriptions, materials, and various engineering parameters. There are also smaller embedded images of spacecraft components within the spreadsheet interface.

[Click](#) for Solar Array Deployment Animation

